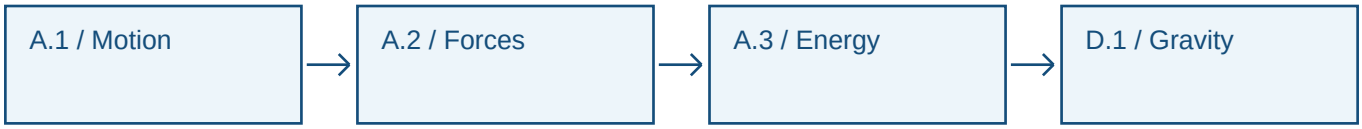
Physics prerequisite roadmap

A content-based analysis of all five themes and 24 subtopics. No example teaching route is used.

How to read the deductions

Core foundations are inferred from the physics needed to understand a topic. A core permission allows earlier content in a query; it does not mean every individual question requires the whole foundation topic. Conditional links apply only to specific extensions and never expand a topic query automatically.

Foundation -> dependent topic



Gravity may include earlier circular motion and energy. A forces query does not gain permission for gravity. Gravity also has a direct forces foundation; the diagram shows the transitive chain.

Dependency layers across the whole syllabus

Layer	Topics with this longest prerequisite depth
0	A.1
1	A.2, A.5
2	A.3
3	A.4, B.1, B.5, C.1, D.1, D.2
4	B.2, B.3, C.2, D.3, D.4
5	B.4, C.3, C.5, E.1
6	C.4, E.2, E.3
7	E.4, E.5

Layers are a topological view of the inferred graph, not a prescribed teaching order. Topics in one layer are not prerequisites for one another. Units, algebra, graphs, proportionality and vector skills underpin the map without being separate topic codes.

A. Space, time and motion

Describe motion, explain it through forces and conservation, and extend to rotating bodies and reference frames.

A.1 Kinematics [SL / HL common]

Position, displacement, velocity and acceleration; motion graphs and constant-acceleration/projectile descriptions.

Core: No earlier topic; foundational skills only. Conditional: None assigned.

Deduction: Entry foundation: algebra, vectors, units and graphs are skills rather than additional syllabus topics.

Evidence: TSM printed pp. 2, 5, 13. Overview plus standard kinematics concepts inferred from the topic title.

A.2 Forces and momentum [SL / HL common]

Newton laws, free-body diagrams, equilibrium, momentum and impulse; collisions and centripetal force for circular motion.

Core: A.1. Conditional: A.3.

Deduction: Acceleration and velocity underpin dynamics. Energy analysis of elastic/inelastic collisions is a conditional extension, avoiding an A.2/A.3 cycle.

Evidence: TSM printed pp. 3, 6, 17. Explicit conceptual overview; impulse is a standard momentum concept.

A.3 Work, energy and power [SL / HL common]

Work by a force, kinetic and potential energy, conservation, power and efficiency.

Core: A.2. Conditional: None assigned.

Deduction: Work requires force and displacement; mechanical energy builds on motion and dynamics.

Evidence: TSM printed pp. 3, 6, 17. Explicit work-energy overview; power/efficiency deduced from topic scope.

A.4 Rigid body mechanics [HL only]

Rotational kinematics and dynamics of extended bodies; torque, rotational inertia, rotational energy and angular momentum.

Core: A.2, A.3. Conditional: None assigned.

Deduction: Linear equations and Newton laws supply rotational analogies; energy and momentum supply conservation methods.

Evidence: TSM printed pp. 3, 6, 13, 17. Rotational analogies/energy are explicit; torque/inertia are standard deductions.

A.5 Galilean and special relativity [HL only]

Reference frames, Galilean comparisons and special-relativistic space-time; time dilation and length contraction.

Core: A.1. Conditional: None assigned.

Deduction: Comparing motion across frames requires velocity, displacement and time. Nuclear mass-energy use does not require the entire relativity topic.

Evidence: TSM printed pp. 2, 5. Time dilation/length contraction explicit; frame comparisons inferred from title.

B. The particulate nature of matter

Use microscopic particles and energy transfer to explain thermal behaviour, gases, atmosphere and circuits.

B.1 Thermal energy transfers [SL / HL common]

Particle models, temperature and internal energy; heat capacity, latent heat, conduction, convection and radiation.

Core: A.2, A.3. Conditional: None assigned.

Deduction: Particle motion/forces and kinetic/potential energy explain microscopic storage and macroscopic transfer.

Evidence: TSM printed pp. 3·6, 9·10. Explicit conceptual overview and prior-knowledge example.

B.2 Greenhouse effect [SL / HL common]

Earth energy balance, absorption and emission of radiation, greenhouse gases and warming.

Core: B.1. Conditional: C.1, C.4.

Deduction: Thermal transfer/radiation are core. Molecular resonance is an explanatory extension, not a prerequisite for every energy-balance problem.

Evidence: TSM printed pp. 4·5, 17. Explicit greenhouse/energy-budget description; resonance treated conditionally.

B.3 Gas laws [SL / HL common]

Ideal-gas model and gas laws; pressure from particle collisions, absolute temperature and molecular kinetic/internal energy.

Core: B.1, A.2. Conditional: None assigned.

Deduction: Thermal variables combine with force/momentum and particle motion to connect microscopic and bulk behaviour.

Evidence: TSM printed pp. 4·6, 9·10. Explicit gas model and kinetic-theory discussion.

B.4 Thermodynamics [HL only]

First and second laws, heat/work/internal energy, gas cycles, heat engines and Carnot efficiency.

Core: B.3, A.3. Conditional: None assigned.

Deduction: Gas states and energy accounting underpin cyclic engines. Entropy is a standard second-law concept, not independently detailed here.

Evidence: TSM printed pp. 4, 6, 17. Laws, cycles and efficiency explicit; entropy detail not established by this source.

B.5 Current and circuits [SL / HL common]

Charge flow and electric current, potential difference as energy per charge, resistance, circuits and electrical power.

Core: A.3. Conditional: B.1.

Deduction: Work/energy and rate concepts define voltage and power. Microscopic temperature-resistance models may add thermal content.

Evidence: TSM printed pp. 4·6. Explicit charge flow/resistance/voltage; circuit relations inferred from title.

C. Wave behaviour

Build from oscillators to propagation, superposition, resonance and motion-dependent observations.

C.1 Simple harmonic motion [SL + additional HL]

Restoring-force oscillations, displacement and acceleration, kinetic/potential energy exchange and damping.

Core: A.2, A.3. Conditional: None assigned.

Deduction: Dynamics defines the restoring acceleration; energy accounts for oscillation and damping.

Evidence: TSM printed pp. 4·6, 17. Explicit oscillation, restoring-force and energy description.

C.2 Wave model [SL / HL common]

Progressive-wave model; amplitude, wavelength, frequency and speed; mechanical, sound and electromagnetic waves and energy transfer.

Core: C.1. Conditional: None assigned.

Deduction: The source explicitly builds the wave model on oscillation. Electromagnetic waves need not import a full fields course.

Evidence: TSM printed pp. 4·5. Oscillation foundation and wave types explicit; basic wave quantities inferred.

C.3 Wave phenomena [SL + additional HL]

Superposition and interference; wave behaviours including reflection, refraction and diffraction.

Core: C.2. Conditional: None assigned.

Deduction: Wavelength, phase and propagation from the wave model underpin overlapping waves and propagation at boundaries.

Evidence: TSM printed pp. 2, 12, 14·15. Superposition/interference explicit; other standard wave phenomena are physics deductions.

C.4 Standing waves and resonance [SL / HL common]

Standing-wave formation and patterns, harmonics, resonance, driving and damping; vibrating strings and air columns.

Core: C.1, C.3. Conditional: None assigned.

Deduction: Superposition creates standing patterns; oscillator response explains resonance.

Evidence: TSM printed pp. 4·6. Standing waves, driving/damping and musical systems explicit; harmonics inferred.

C.5 Doppler effect [SL + additional HL]

Observed frequency/wavelength changes from source-observer relative motion; sound/light Doppler interpretation and redshift.

Core: C.2, A.1. Conditional: A.5.

Deduction: Wave quantities and relative velocity are core. Special-relativistic treatment is conditional and must be separately in scope.

Evidence: TSM printed pp. 2. Topic name supports a physics deduction; detailed Doppler coverage is not provided in this overview.

D. Fields

Model interactions at a distance and apply force/energy reasoning to masses, charges and induction.

D.1 Gravitational fields [SL + additional HL]

Universal gravitation, field strength and field maps, orbital/circular motion and Kepler relations; potential, energy and escape.

Core: A.2, A.3. Conditional: None assigned.

Deduction: Gravity acts as a force; circular motion uses dynamics; potential and escape use work-energy.

Evidence: TSM printed pp. 4, 6, 11, 17. Explicit overview and progression-of-learning example; some content is additional HL.

D.2 Electric and magnetic fields [SL + additional HL]

Electric and magnetic field models; Coulomb forces and field lines; electrical potential energy and potential gradients.

Core: A.2, A.3. Conditional: D.1, B.5.

Deduction: Force and work-energy supply definitions. Gravity analogy and current-generated field/circuit applications are conditional.

Evidence: TSM printed pp. 4, 6, 17. Explicit force/field/potential descriptions; topic combines SL and additional HL.

D.3 Motion in electromagnetic fields [SL / HL common]

Charged-particle trajectories in electric/magnetic fields, circular/parabolic paths, combined fields and specific-charge measurements.

Core: D.2, A.2, A.3. Conditional: B.5.

Deduction: Field forces and Newtonian trajectories combine; electric acceleration uses energy. Circuit/beam-source apparatus is conditional.

Evidence: TSM printed pp. 5·6, 11·12, 17. Explicit conceptual overview and assessment-learning example.

D.4 Induction [HL only]

Changing magnetic flux and induced emf/current; Faraday/Lenz reasoning, electricity generation and energy conservation.

Core: D.2, B.5, A.3. Conditional: None assigned.

Deduction: Magnetic field/flux, circuit quantities and work-energy are foundations; Lenz direction enforces conservation.

Evidence: TSM printed pp. 4, 17. Lenz/generation explicit; flux and Faraday relation are standard induction deductions.

E. Nuclear and quantum physics

Connect atomic/nuclear structure, quantization, radiation and nuclear energy with microscopic evidence and stellar observations.

E.1 Structure of the atom [SL + additional HL]

Rutherford nuclear atom, electrons/protons/neutrons; Bohr quantized levels and absorption/emission spectra.

Core: D.2, C.2, A.3. Conditional: None assigned.

Deduction: Scattering/closest approach use field forces and energy; spectra connect photon energy with wave frequency.

Evidence: TSM printed pp. 4·5, 12, 17. Explicit overview and atom-model example; mixed SL/additional HL depth.

E.2 Quantum physics [HL only]

Photoelectric energy balance, wave-particle duality, de Broglie matter waves and Compton scattering.

Core: E.1, C.2, A.2. Conditional: B.5, C.3.

Deduction: Atomic/photon energy and wave quantities combine with momentum. Stopping-potential circuits and diffraction evidence are conditional.

Evidence: TSM printed pp. 4, 6, 17. Photoelectric/de Broglie/Compton explicit; experimental prerequisites deduced.

E.3 Radioactive decay [SL + additional HL]

Nuclear composition/stability, short-range nuclear force, radioactive transformations, decay/half-life and binding-energy mass changes.

Core: E.1, A.3. Conditional: None assigned.

Deduction: Atomic nuclear structure and conservation/energy underpin nuclear change; the whole A.5 course is not necessary for a supplied mass-energy relation.

Evidence: TSM printed pp. 4, 7, 17. Nuclear forces/binding energy explicit; decay/half-life are standard deductions from title.

E.4 Fission [SL / HL common]

Fission of heavy nuclei, chain reactions, nuclear energy release and nuclear-power generation.

Core: E.3. Conditional: B.1, B.4.

Deduction: Binding-energy and nuclear-reaction ideas are core. Thermal plant operation and engine efficiency are conditional.

Evidence: TSM printed pp. 4, 17. Fission/power and mass-energy explicit; chain reactions inferred from content scope.

E.5 Fusion and stars [SL / HL common]

Fusion as a stellar energy source, stellar observations/spectra and thermal radiation; temperature, luminosity and stellar models.

Core: E.3, B.1. Conditional: D.1, B.3, C.5, E.4.

Deduction: Nuclear energetics and thermal radiation are core. Gravitational equilibrium, gas models, Doppler and comparison with fission are conditional.

Evidence: TSM printed pp. 4, 10, 17. Fusion/stars, spectra, brightness, temperature and radius explicit. This is not an exhaustive astrophysics syllabus.

Rules for topic-safe selection

Use the content prerequisites together with a separately approved subquestion dependency map.

1. Expand only core foundations

Allowed scope = selected targets plus their transitive core foundations. Conditional links are excluded unless selected explicitly or already reached through a core path. Never infer reverse links or all same-theme topics.

2. Require a target task

At least one retained task must be tagged with a selected target. A mechanics-only question is not a gravity result, even though mechanics is in the gravity knowledge scope.

3. Remove out-of-scope extensions

For A.1 motion, remove an optional A.3 energy extension. For D.1 gravity, allow A.2 circular motion; for A.2 forces, do not permit D.1. An inseparable multi-topic task needs all its topics in the allowed scope.

4. Preserve the dependency direction

If X requires A, retaining X always retains A and the shared stem. Retaining A does not force X. When an out-of-scope prerequisite is removed, remove its dependants too. Reject the parent if no target task survives or the shared stem is out of scope.

5. Keep approval and level checks

Only current, reviewed part crops, topic tags and acyclic dependency maps authorize trimming. Missing or stale maps permit a whole question only if all parent tags fit. HL/SL suitability is a separate check; mixed-depth topic codes do not certify SL content.

6. Apply Full and Short consistently

Unfiltered Full preserves every crop. Topic-filtered Full keeps every in-scope task. Short may additionally omit optional tasks, preferring independent tasks, while preserving a target task and all required context.

Source and limits

IB Physics teacher support material, first assessment 2025; published September 2023, updated July/November 2024. Analysis uses printed pp. 2-17 (PDF pp. 7-22); the sample teaching route is excluded. The uploaded PDF has 116 pages. Full source identity and per-topic rationales are in curriculum_prerequisites.json.

This is a content-based project synthesis, not an official IB prerequisite specification or an exhaustive syllabus checklist. The support material is brief on some topics; the cards label physics deductions where details are inferred. Core versus conditional is an application judgement that can be revised after teacher review.

Small-model dependency suggestions require human approval. Current real inference has not demonstrated reliable dependency classification; conservative validation rejects invalid graphs.